Introduction to memories

M. Sonza Reorda

Politecnico di Torino
Dip. di Automatica e Informatica



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Introduction

Each computer system includes units for storing data and instructions.

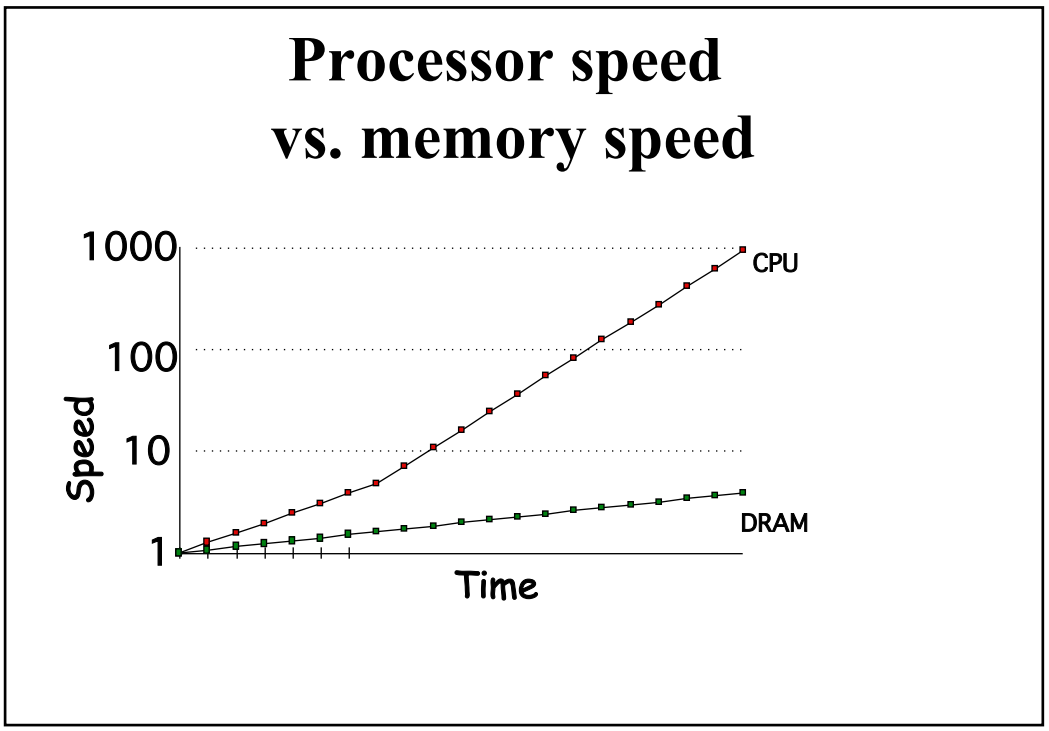
All of these units, together with the algorithms for their management, compose the *memory sub-system*.

This sub-system must be designed so that:

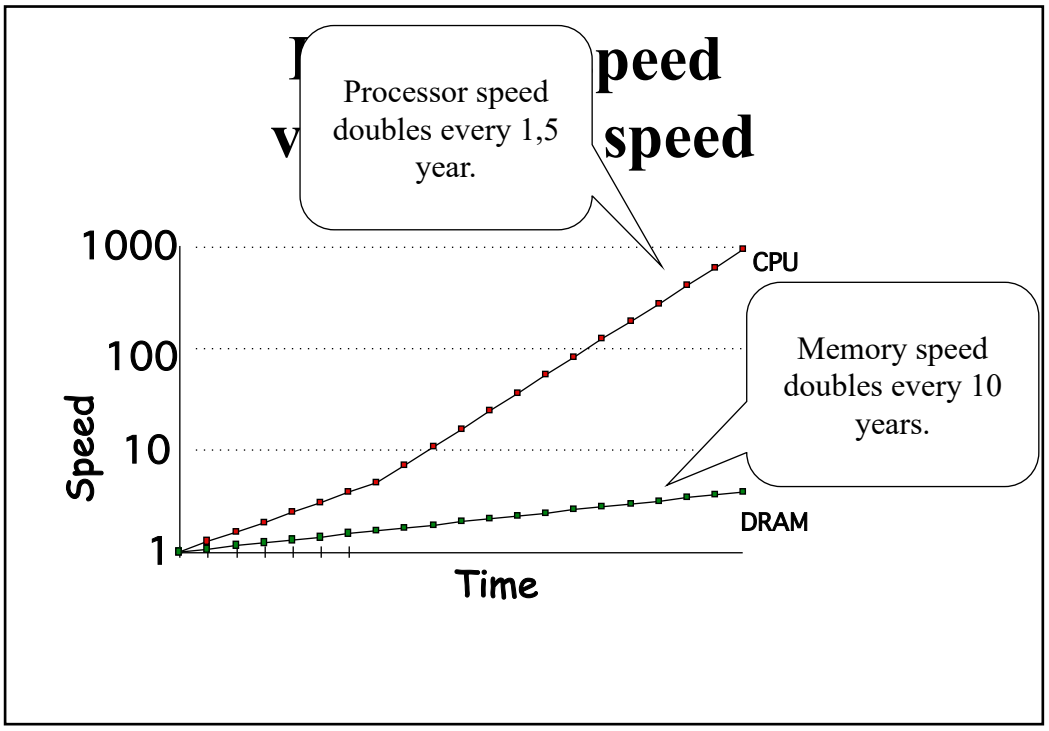
- **the processor must wait as little as possible to access data or instructions**
- **the cost per bit of the sub-system is minimal (taking into account that the cost of memory is proportional to their speed).**

A compromise must be found between the cost of the memory sub-system and its performance.

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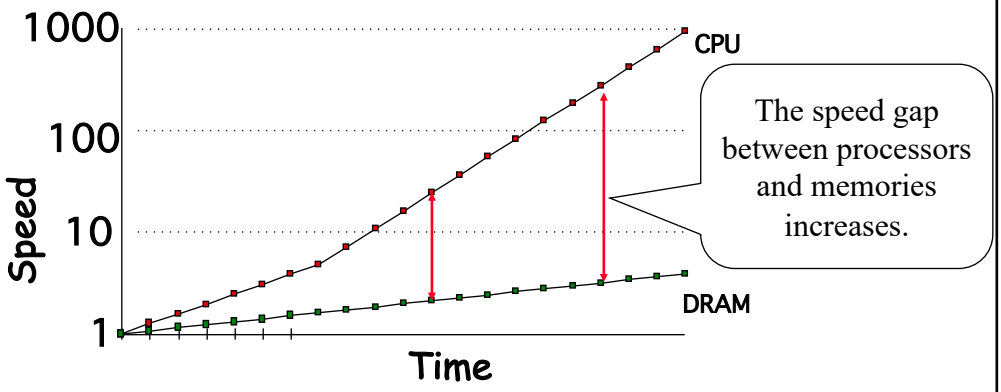

Processor speed
vs. memory speed
1000
100
10
1
Speed
Time
CPU
DRAM

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Processor speed doubles every 1,5 year.
Memory speed doubles every 10 years.
1000
100
10
1
Speed
Time
CPU
DRAM

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Processor speed vs. memory speed

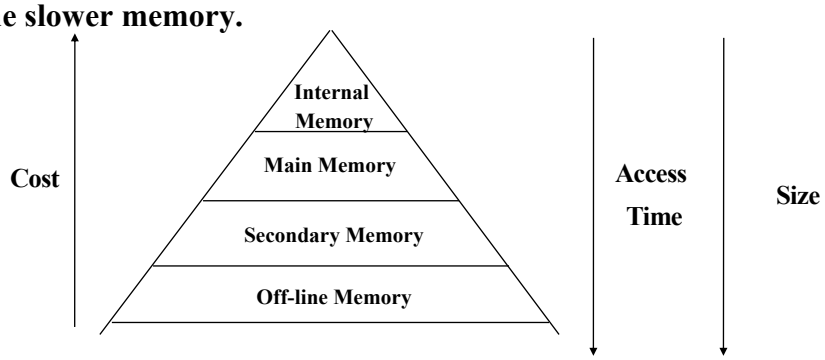


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Memory levels

The memory of a computer is usually organized in levels. Each level corresponds to a different type of memory (in terms of speed, cost and size).

In this way you can achieve performance comparable with those of the faster memory at a price comparable with that of the slower memory.



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Processor internal memory

It corresponds to the internal registers of the processor.

It is characterized by:

- **high speed (the same of the processor): the access times is in the order of a few ns**
- **small size (at most a few thousand bytes).**

It is accessible directly by name.

It is usually implemented as *static RAM cells*.

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Main memory

- **It is much larger (up to a few GiBytes), but its access time is higher (in the order of tens of ns).**
- **It is accessible by address.**
- **It is usually implemented as *dynamic RAM cells*.**
- **Information stored in the primary memory are lost when the memory is powered off.**

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Secondary memory

- **Its size and access time are larger than for the main memory (in the order of tens of ms).**
- **It is used to store data and programs which are not immediately required by the processor.**
- **Its access is managed by special interface programs.**
- **It is normally implemented in the form of *magnetic disks* or *flashes*.**
- **Information stored in the secondary memory are not lost when the memory is powered off.**

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Off-line Memory

- **It allows you to store large amounts of data (from GiB to TiB) with high access times (tens of seconds).**
- **In most cases, access to off-line memory requires human intervention.**
- **It is usually composed of *optical disks* or *tapes*.**

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Cache

- **Caches are very fast memories which are interposed between the processor and main memory.**
- **A cache temporarily stores data / programs used at that time.**
- **Their use is transparent to the programmer and the operating system.**
- **The cache can improve the speed of data access in main memory.**

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General strategy

When designing a memory sub-system the following points must be taken into account:

- **the overall system should be composed of memories of different types and costs (*memory hierarchy*)**
- **memory management must be as transparent as possible to the programmer and the user (*virtual memory*)**
- **if the system is part of a multiprocessor system, each processor must be able to work with the memory at full speed and without interfering with the work of the others.**

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Memory classification

It is based on various parameters:

- **cost**
- **speed**
- **access modes**
- **alterability**
- **durability**
- **reliability**
- **physical characteristics.**

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Cost

- **It also includes the cost of the circuitry and possibly the of the software needed to manage the memory.**
- **Is normally measured in $/bit o $/MiByte.**

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Speed

It is expressed via 3 parameters

- **The access time**
- **The cycle time**
- **The transfer rate.**

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Access Time

This is the time between the instant when the request for an access (read or write) reaches the memory unit, and that in which the operation is performed.
It is possible that the access time for read operations is different than the one for write operations.
The access time is usually inversely proportional to the cost of the memory.

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Cycle time

- **It is the time between the beginning of one access and the beginning of the following.**
- **It is largely or equal to the access time.**

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Transfer rate

- **It is the speed of the data transfer to / from the memory unit.**
- **For the random access memories it is the inverse of the cycle time.**
- **For the others**

$$T_N = T_A + n/R$$

where

- T_N **is the average time to read / write** n **bits**
- T_A **is the average access time**
- n **is the number of bits**
- R **is the transfer rate (in bit per second, or** *bps***).**

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Memory evolution

The technology evolution has two trends:

- **reducing the *cost per bit***
- **reducing (more slowly) the *access time*.**

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Access Modes

The main ones are:

- ***Sequential mode*: information can be read / written only in a fixed order; it is the case of optical memories and tapes.**
- ***Direct mode*: each block has an address, and search in the block is sequential; it is the case of magnetic disks.**
- ***Random mode*: each data has a given address, and information can be read / written in any order by specifying its address; the access time is equal for all memory locations; it is the case of semiconductor memories.**
- ***Associative mode*: access is done by comparing the content of each cell with the one specified in a mask; it is the case of certain types of caches.**

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Alterability

It is the property of memories whose content can be modified during their normal behavior.

There are memories whose contents can be written only once (Read Only Memory, or ROM).

In some cases the contents of a ROM can be edited off-line (Programmable ROM, or PROM).

In other cases, the memory content can be modified at any time (as for Random Access Memories, or RAM).

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Durability: Destructive Readout

There are some special memory technologies where the read operation causes the deletion of the stored data (*Destructive Readout*, or DRO).
In this case, after each read operation it is necessary to perform a write operation with the same data.

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Durability: refreshing

- In some technologies (such as dynamic RAMs) after some time cells storing a 1 change their content into a 0.
- This effect occurs for example when the bit is stored as a charge in a capacitor, because of the leakage current.
- Therefore, it is necessary to perform periodically a read and re-write on every cell (*refresh*).

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Durability: volatility

Some memories, such as semiconductor ones, lose their content when they are powered off (*volatile memories*).

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Reliability

- **Memories can be affected by two types of faults:**
  - ***Transient faults*: one or more bits suddenly change their value, but the memory still continues to work correctly**
  - ***Permanent faults*: something in the memory permanently stops working.**
- **Reliability is usually measured via the following parameters:**
  - ***Mean Time To Failure* (MTTF): mean time before a failure occurs**
  - ***Failure rate (or Mean Time Between Failures,* MTBF)*:* average frequency of occurrence of failures.**

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Other characteristics

- **Memory type (electronic, mechanical, magnetic, optical)**
- **Power consumption**
  - **It may cause the need for cooling systems**
  - **It can be critical for portable systems**
- **Portability**
- **Storage density**
- **…**

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